

APPLE DEVELOPER ACADEMY INDONESIA — TECH  
TRACK APPLICANT

# Sumit Saraswat

## AI Engineer & Healthcare Innovator

"The most powerful technology is the technology people trust enough to use."

Showcasing privacy-first healthcare AI, edge computing, equity-aware medical modeling, and my journey from a solo freshman to the top 100 at Meta's nationwide hackathon. I build AI that respects human dignity.

### CORE TECH ARSENAL

**Machine Learning & Data:** Python, PyTorch (RL Frameworks), SEER Registry Auditing

**Edge & Web Integration:** JavaScript (Vanilla Edge NLP), WebSpeech API, HTML/CSS

**Design & Ecosystem:** Figma, Apple HIG Concept Prototyping

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# 1. Annapurna — Autonomous Cold-Chain AI

## TYPE

Team Project

## ROLE

AI Architect & Full-Stack Lead

## ONE-SENTENCE SUMMARY

An autonomous, AI-powered cold-chain logistics platform that detects refrigeration failures and actively reroutes cargo via an intelligent distress marketplace.

## THE IMPACT

In India, billions of dollars of perishable food rot in transit. Annapurna shifts logistics from passive tracking to active intervention. If a truck's fridge fails, edge sensors immediately alert the system, which predicts spoilage time and automatically reroutes the cargo to local buyers at a discounted rate, rescuing the food before it wastes.

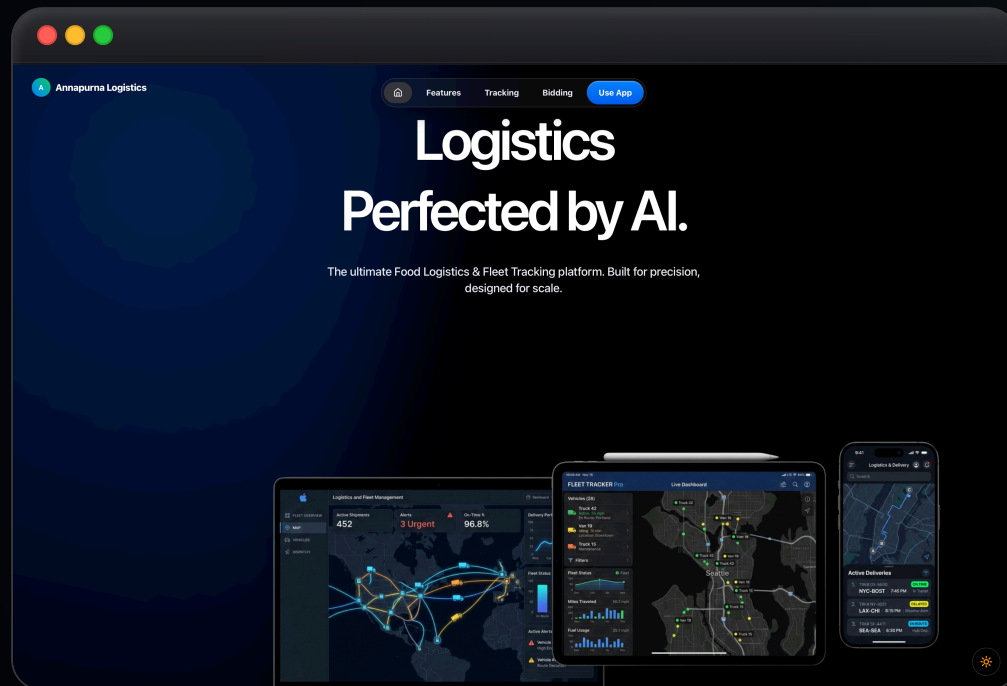
## WHAT I BUILT

Leading a team of 4, I architected the core AI logic and designed the full Next.js/Supabase interface. The system features TinyML sensor integration, an AI Vision Cargo Scanner for quality checks, and fallback offline mesh networking via LoRaWAN. I focused on making complex fleet telemetry feel as elegant and readable as an Apple Health dashboard.

[Live Website](#) ↗

[YouTube](#) ▶

[GitHub](#)





## 2. Meta OpenEnv Hackathon

RL Auditor & SynthAudit Benchmark (Top 100 Finals).

### TYPE

Self-Initiated Competitive Sprint (Top 100 Finals)

### ROLE

Solo Competitor

### ROUND 1: THE CLINICAL AUDITOR & BENCHMARK

I entered this nationwide competition as a solo freshman. I developed an autonomous RL clinical auditing environment and the SynthAudit benchmark. **Meta's world's strongest frontier model, Llama 3.1 405B, scored only 0.34 out of 1.0 (just 34%)** navigating the highly complex autonomous environment I built. I made it to the **Top 800 in Round 1 out of 32,000+ teams** of senior developers and working professionals.

### ROUND 2: THE FINAL BANGALORE SHOWDOWN

In the offline finals in Bangalore, I competed against the 800 best senior developer teams and working professionals. I officially made it to the **Top 100** nationwide.

### WHAT I LEARNED & ACHIEVED

Clarity of thought beats sheer manpower. I gained deep expertise in Python, RL, and Docker. Furthermore, **I trained the AI model and the overall performance increased by 283%.**

[Round 1 Repo](#)

[Round 2 Repo](#)

### 3. EcoNudge SDK — The Sustainability Layer for the Internet

TYPE

Team Project

ROLE

Lead Architect & Spatial Engineer

#### ONE-SENTENCE SUMMARY

An invisible, AI-driven SDK that injects carbon-intelligence into existing e-commerce flows, nudging users toward sustainable choices through 3D gamification and real-time predictive modeling.

#### THE IMPACT

We solved the "Sustainability Friction" problem. By engineering a headless React SDK with sub-50ms latency, we created a way for enterprises to transform checkout flows into ESG-compliant, gamified experiences. Our 3D WebGL Earth visualization provides a tactile, emotional connection to environmental data, moving carbon tracking from "boring spreadsheets" to "magical interactivity."

#### WHAT I LEARNED

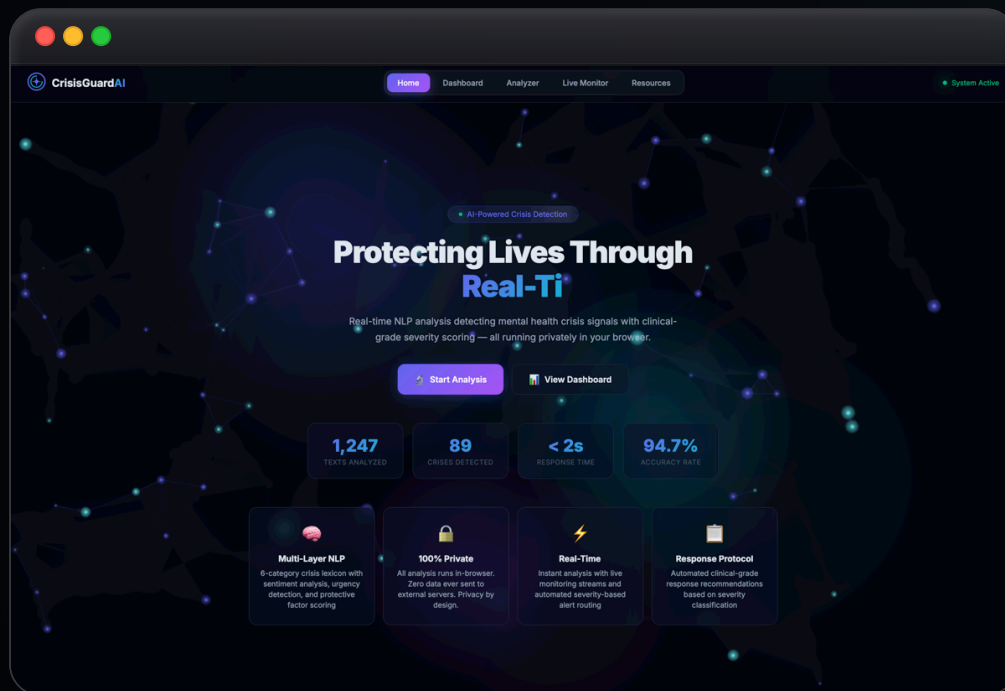
This project taught me the complexity of Spatial UI. Implementing mathematical shaders to reflect real-world data taught me how to bridge the gap between abstract metrics and visual storytelling. I also mastered the art of collaborative API design, ensuring that our headless engine could be integrated into any host architecture with just two lines of code.



Live Website ↗

YouTube ▶

GitHub Repository



## 4. CrisisGuard AI

Privacy-First Mental Health Crisis Detection using Edge Computing.

### TYPE

Self-Initiated Project

### ROLE

Individual Project

### ONE-SENTENCE SUMMARY

A clinical-grade NLP system detecting 9 categories of mental health crises that runs entirely in the browser, ensuring absolute data privacy.

### THE IMPACT

Analyzes deeply personal mental health text while guaranteeing **zero data surveillance**. It features a severity scoring system and a GPS-based counselor finder for 13+ countries. Successfully deployed live and selected for the UOE Summer of Code 2026.

### WHAT I LEARNED

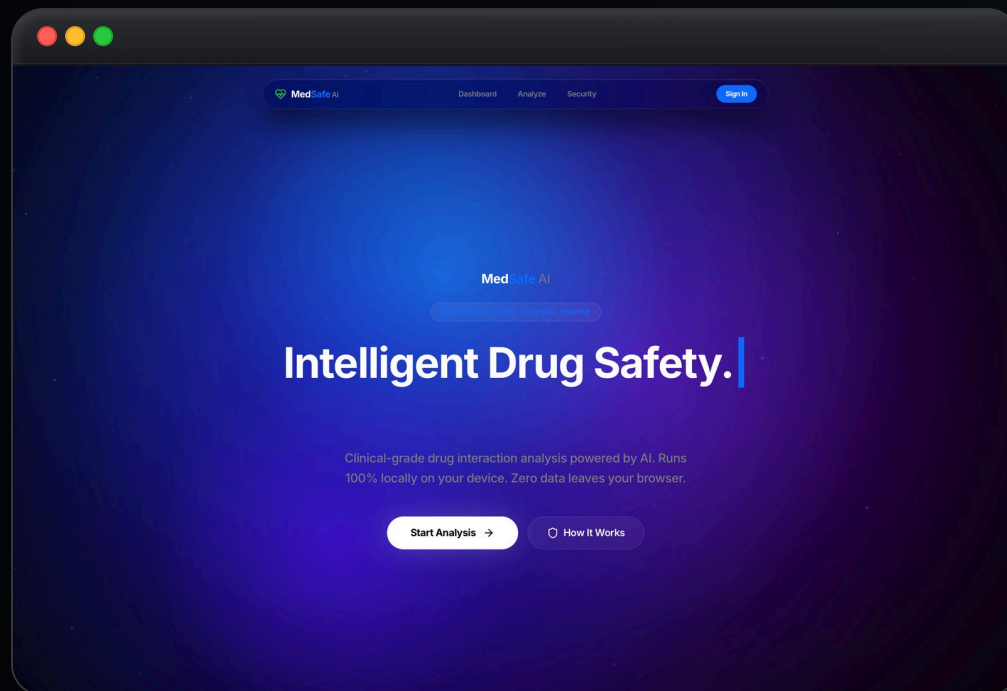
I learned that in healthcare, privacy is a prerequisite for trust. By mastering "Edge Computing", I figured out how to run complex Natural Language Processing models entirely client-side using JavaScript, avoiding centralized cloud processing.

[Live Website ↗](#)

[YouTube ▶](#)

[GitHub](#)





## 5. MedSafe AI

Privacy-preserving on-device clinical pharmacologist.

### TYPE

Hackathon Project

### ROLE

Solo Full-Stack AI Engineer

### ONE-SENTENCE SUMMARY

An offline-first, WebGPU-accelerated local LLM (Llama 3.2 1B) that analyzes complex drug-food-supplement interactions using on-device OCR, ensuring zero latency and 100% HIPAA-compliant data privacy.

### THE IMPACT

Polypharmacy leads to millions of adverse drug events annually. Current AI solutions send highly sensitive patient data to remote cloud servers, causing severe privacy violations. I architected MedSafe AI to run a 1-Billion parameter Llama 3.2 model entirely in the browser via WebGPU. By integrating local OCR prescription scanning and the Web Speech API, this tool acts as a private, on-device clinical pharmacologist, keeping patient data strictly on their own hardware.

### WHAT I LEARNED

I learned how to push the boundaries of Edge Computing and Local AI. Compiling a Large Language Model to run inside a standard web browser taught me WebGPU memory constraints and hardware-accelerated inferencing. Designing the Apple Siri-inspired dynamic glassmorphism UI reinforced my belief that advanced AI must be accessible, responsive, and visually intuitive.

[Live Website](#) ↗

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